

Punishment or Pruning? Diagnosing Algorithmic Collusion among Reinforcement-Learning Traders

Hieu Tran
University of Massachusetts Amherst
hieutran@umass.edu

Abstract

Q-learning informed traders in Kyle markets trade far less than competition implies, earning supracompetitive profits and making prices less informative. Whether this is *collusion*, sustained by the threat of punishment, or a *learning bias* that prunes aggressive orders matters for policy: the first calls for limits on transparency, the second for standards on algorithm design. We separate the two with a paired rival-deviation test and three placebos that cannot collude: myopic learners, an uninformative memory, and passive traders that move the cartel above Nash. The same diagnostics find punishment in the pricing game of Calvano et al. (2020), so they can fail. Analytically, a one-period deviation in the standard Kyle market moves order flow by only $(I-1)/(2I)$ noise standard deviations per unit of the value signal, whatever the noise volume, and Nash-reversion trigger strategies cannot sustain any collusion; with many information-insensitive investors deviations are hundreds of noise standard deviations large. In simulations, Q-learners close 0.86 of the gap from competition to the cartel, yet never punish deviations, even with perfect monitoring, and deviating pays. Myopic learners trade even less (1.24), an uninformative memory reproduces the outcome, and with passive traders learners trade half their Nash intensity although collusion would mean trading more. A lone trader with no rival under-trades by up to 57%, in proportion to the square root of the step size. Reducing transparency changes nothing (0.89); counterfactual updating restores competition (-0.10). Where deviations are hundreds of noise standard deviations large, forward-looking learners still never punish, while a myopic shared-table learner shows the reactions to price shocks and deviations that have been read as punishment. We turn the diagnostics into an audit protocol and release a validated, CPU-only testbed with exact benchmarks, tabular and deep RL agents, and all diagnostics.

1 Introduction

Trading is increasingly delegated to algorithms that learn from their own profits. A growing literature shows that such algorithms can arrive at *supracompetitive* outcomes without communicating: reinforcement-learning (RL) pricing agents settle on prices above the competitive level [7, 23], algorithmic market makers quote wider spreads [10, 11], and RL informed speculators trade less aggressively than competition would imply, earning collusive profits and degrading price informativeness [14]. For regulators the finding is alarming, because tacit collusion by algorithms falls outside the reach of rules written around agreements among humans [18, 22].

A supracompetitive outcome, however, is not the same thing as collusion. In the game-theoretic sense, collusion is an outcome sustained by the threat of punishment: a trader who deviates to grab a larger share of the rent is punished by its rivals, so deviating does not pay. The same low-intensity outcome can instead come from a learning bias: an algorithm that systematically undervalues aggressive actions will trade conservatively even when no rival could ever punish it. Dou et al. [14]

name both channels in the trading context (“price-trigger” collusion and “over-pruning” bias), and a parallel debate in industrial organization asks whether Q-learning collusion is genuine or an artifact of exploration and learning design [1, 3, 5, 8, 12]. The distinction is not academic. If algorithms punish, transparency matters, because monitoring is what makes punishment possible, and the natural remedies target information: reporting delays, reduced order-flow disclosure. If the outcome is a learning bias, transparency is irrelevant and the remedies target algorithm design.

This paper asks how to tell the two apart, and what the answer is for RL informed traders in the canonical Kyle [25] market and in the Dou et al. [14] extension with information-insensitive investors.

Contributions.

1. **Diagnostics that can fail, and do.** We combine a paired *rival-deviation* test (one agent is forced to play its one-period best response; we measure the rivals’ reaction under common random numbers) and the Dou et al. [14] *noise-shock* test with placebos that cannot collude: a *myopic* learner ($\gamma = 0$), *no memory*, an *uninformative memory* of the same size as an informative one, and *passive traders* that move the collusive outcome above the Nash one, so that collusion and bias predict opposite directions. Applied to the logit Bertrand game of Calvano et al. [7], the diagnostics find punishment (the rival cuts its price and deviating loses), and they flag the same game without memory, where prices are even higher ($\Delta = 0.95$), as a supracompetitive outcome with no punishment behind it.
2. **When can traders punish at all?** In the standard Kyle market a one-period deviation from the collusive intensity moves order flow by only $(I - 1)/(2I)$ noise standard deviations per unit of v/σ_v , whatever the volume of noise trading; with a large mass of information-insensitive investors the ratio is about 625 in the Dou et al. [14] calibration (Proposition 2). Building on Green and Porter [21], we show that in the standard market Nash-reversion trigger strategies cannot sustain *any* collusion, so a near-cartel outcome would need much harsher punishments, and we bound what any punishment scheme can sustain (Propositions 3 and 4). A myopic learner whose estimates are unbiased plays Nash (Proposition 5), so the myopic placebo measures learning bias directly.
3. **The bias, isolated.** A single informed trader with no rival and a fixed price rule under-trades by up to 57% under standard Q-learning, by an amount proportional to $\sqrt{\alpha}$ ($R^2 = 0.994$), while counterfactual updating learns the optimum.
4. **Low trading without collusion.** In the standard Kyle market, forward-looking Q-learners close on average 0.86 of the gap between competition and the cartel, yet rivals never punish deviations, even with perfect monitoring, and deviating pays. The myopic placebo trades even less ($\Delta = 1.24$, beyond the cartel); an uninformative memory reproduces the outcome (0.84); and when passive traders put the cartel *above* Nash, learners still trade half their Nash intensity and earn 30% less than in equilibrium. Blurring the order flow changes nothing ($\Delta = 0.89$), while counterfactual updating, a change to the learning rule alone, restores competition ($\Delta = -0.10$).
5. **Where deviations are glaring.** In the Dou et al. [14] regime, forward-looking learners still never react to shocks or deviations, while a *myopic* shared-table learner reacts to both: it shows the threshold-shaped response to price shocks that has been read as a price trigger, although it has no reason to punish and deviating still pays. These signatures do not identify collusion without a placebo.

6. **An audit protocol and an open testbed.** We assemble the diagnostics into a five-step protocol that returns *collusion* for the Calvano et al. [7] baseline and *learning bias* for every market of informed traders we study, and apply it to competing Q-learning dealers under adverse selection. We release a batched, CPU-only simulator with exact benchmarks, tabular Q-learning, DQN and PPO agents, a compiled training engine, all diagnostics, and a test suite that validates the economics and the diagnostics against scripted agents whose behaviour is known.

Why this matters for policy. The two channels call for different interventions. Our results suggest that in markets like the standard Kyle setting, the supracompetitive outcomes produced by off-the-shelf Q-learning are properties of the learning procedure, and would respond to algorithm-design standards (step sizes, exploration) rather than to limits on transparency. Where deviations are highly detectable, as with a large mass of information-insensitive investors, the diagnostics must be applied with a placebo, because a reaction to a price shock is not by itself evidence of punishment.

2 Related Work

Algorithmic collusion. Calvano et al. [7] show that Q-learning agents in a repeated Bertrand game learn supracompetitive prices sustained by punishment, documented through impulse responses to a forced deviation; Klein [23] obtains similar results under sequential pricing. A second wave questions how these outcomes arise. Asker et al. [3, 4] show that the outcome depends on algorithm design: asynchronous updating (learning only from the action taken) yields near-monopoly prices, while synchronous, counterfactual updating yields competitive ones. Abada and Lambin [1] show that algorithms quickly reach seemingly collusive outcomes and that a regulator can restore competition by enforcing decentralised learning or intervening in the learning process. Banchio and Mantegazza [5] show that collusion can arise from spontaneous coupling of the learning dynamics rather than from punishment strategies. den Boer et al. [12] argue that the collusive equilibria of Q-learning are reached only on time scales irrelevant in practice and require implausibly synchronized algorithms. Calvano et al. [8] discuss when algorithmic collusion is genuine or spurious. Deep RL agents [13] and large language models [19] can also price above competitive levels. We contribute a placebo-based diagnostic (the myopic learner) and a detectability analysis to this debate, in a financial market with adverse selection.

Algorithms in financial markets. Colliard et al. [10] study Q-learning market makers and find mark-ups over the competitive price driven by limited experimentation and noisy feedback, the same mechanism we find for informed traders. Cartea et al. [9] show that the tick size bounds the excess rents of learning liquidity providers, and Cont and Xiong [11] study tacit collusion among market-making algorithms in dealer markets. Dou et al. [14] extend Kyle [25] to several informed speculators, information-insensitive investors, and a market maker who trades off pricing error against inventory. They show that Q-learning speculators sustain collusive profits and reduce price informativeness, and separate price-trigger collusion from over-pruning bias with noise-shock impulse responses, finding the former only when information-insensitive investors are present. An independent reimplementation [15] reproduces both regimes and finds that the level of collusion depends on the update protocol. Our paper adds the rival-deviation test, the myopic placebo and the learning-rate sensitivity analysis, explains the regime difference analytically, and provides a validated open testbed.

Estimation bias in Q-learning. The max operator makes Q-learning over-estimate action values under noise [30, 31]. The bias relevant here runs the other way for rarely chosen actions: with a constant step size and ε -greedy exploration, an action whose estimate falls below the greedy action by chance is rarely updated again, so pessimistic errors persist while optimistic ones are corrected. Actions with noisier payoffs are more exposed. Decreasing step sizes that satisfy the Robbins and Monro [27] conditions remove the persistent component asymptotically [17].

3 Model

3.1 The repeated market

Time is discrete, $t = 1, 2, \dots$. Each period a fresh asset value v_t is drawn i.i.d. with mean $\bar{v}$ (normalised to 0) and variance σ_v^2 . $I \geq 2$ learning informed traders observe v_t and submit market orders $x_{i,t}$. Noise traders submit $u_t \sim \mathcal{N}(0, \sigma_u^2)$, and $P \geq 0$ optional *passive* informed traders play a fixed Nash strategy. Following Dou et al. [14], a mass $\xi \geq 0$ of information-insensitive investors demands $z_t = -\xi(p_t - \bar{v})$. A market maker observes aggregate informed and noise flow $y_t = \sum_i x_{i,t} + u_t$ and sets

$$p_t = \arg \min_p \theta (p - \mathbb{E}[v_t | y_t])^2 + (y_t + z_t)^2 \implies p_t = \bar{v} + \lambda y_t, \quad \lambda = \frac{\theta \lambda_B + \xi}{\theta + \xi^2}, \quad (1)$$

where $\lambda_B = \text{Cov}(v, y) / \text{Var}(y)$ is the Bayesian price impact. With $\xi = 0$ this is the Kyle [25] market maker. Trader i earns $\pi_{i,t} = (v_t - p_t) x_{i,t}$. The value is revealed at the end of the period, and the one-period game repeats.

3.2 Benchmarks

With linear strategies $x_i = \beta_i v$, profits depend only on second moments, so the benchmarks below hold for any value distribution as long as pricing is linear. Let $n = I + P$.

Proposition 1 (Nash and collusive benchmarks). *Let $\xi = 0$. The symmetric Nash equilibrium of the stage game has $\beta^N = \sigma_u / (\sqrt{n} \sigma_v)$ and $\lambda^N = \sqrt{n} \sigma_v / ((n+1) \sigma_u)$. If the learners jointly maximise their profit against a market maker who prices their joint strategy rationally, while passive traders keep playing β^N (aggregate $B_P = P \beta^N$), the learners' aggregate intensity is*

$$B_L^C = \sqrt{B_P^2 + \sigma_u^2 / \sigma_v^2}.$$

For $\xi > 0$, the Nash intensity solves $\beta = 1 / ((n+1) \lambda(n\beta))$, which has a unique root, and the collusive intensity maximises $B_L \sigma_v^2 (1 - \lambda(B_L + B_P) (B_L + B_P))$.

Collusion lowers aggregate intensity ($B_L^C < I \beta^N$ for $P = 0$), raises learners' profit, and reduces price informativeness $\mathcal{I} = \text{Corr}(p, v)^2 = B^2 \sigma_v^2 / (B^2 \sigma_v^2 + \sigma_u^2)$. Following Calvano et al. [7] we report any outcome metric m as a *collusion index*

$$\Delta_m = \frac{m - m^N}{m^C - m^N},$$

so $\Delta = 0$ is competition and $\Delta = 1$ is the cartel. We use aggregate intensity (the estimated slope of learners' total order on v), informativeness, and profit. Because the Nash–cartel profit gap is only 6% at $I = 2$, intensity and informativeness are the more precise measures.

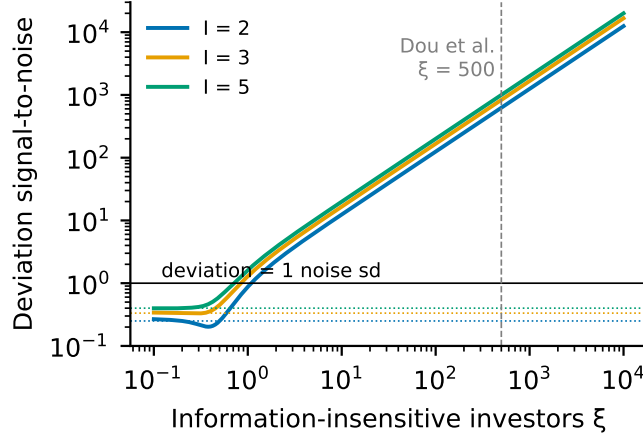


Figure 1: Signal-to-noise ratio of a one-period best-response deviation from the collusive intensity (order-flow change per unit of v/σ_v , in noise standard deviations), as a function of the mass of information-insensitive investors ($\sigma_v = 1$, $\sigma_u = 0.1$, $\theta = 0.1$). Dotted lines: the $\xi = 0$ values $(I - 1)/(2I)$. Deviations become visible above noise only once ξ exceeds about one; at the Dou et al. [14] calibration they are several hundred noise standard deviations large.

Remark 1 (Passive traders can remove the gain from collusion). With $\xi = 0$ and $\sigma_v = \sigma_u$, $B_L^C = \sqrt{P^2/n + 1}$ and $I\beta^N = I/\sqrt{n}$, which coincide exactly when $I = P + 1$. In that case the learners have nothing to gain from colluding and Δ is undefined. When $P > I - 1$ the order reverses: the collusive aggregate *exceeds* the Nash aggregate, because extra informed flow lowers the market maker's price impact, so collusion would mean trading more. This makes passive traders a useful falsification test (Section 5.6).

3.3 How detectable is a deviation?

Punishment requires monitoring. The relevant quantity is how much a deviation moves what traders observe relative to noise. Suppose rivals play the collusive intensity and the market maker prices at λ^C . A trader's myopic best response is $\beta^d = (1 - \lambda^C(I - 1)\beta^C)/(2\lambda^C)$, and a one-period deviation moves aggregate flow by $(\beta^d - \beta^C)v$ against noise of standard deviation σ_u .

Proposition 2 (Deviation signal-to-noise). *Let $P = 0$ and define $\text{SNR} = (\beta^d - \beta^C)\sigma_v/\sigma_u$, the flow change of a best-response deviation per unit of v/σ_v , in noise standard deviations.*

1. *If $\xi = 0$, $\text{SNR} = (I - 1)/(2I) < 1/2$ for all I , independently of σ_u and σ_v .*
2. *As $\xi \rightarrow \infty$ with θ fixed, $\text{SNR} = \xi\sigma_v(I - 1)/(4I\sigma_u)(1 + o(1))$.*

The proof is in Appendix A; Figure 1 plots the exact ratio. In the standard Kyle market every intensity scales with σ_u/σ_v , so changing noise volume rescales deviations and noise alike and cannot make monitoring easier. A single deviation is less than half a noise standard deviation even at $|v| = \sigma_v$, so a trigger strategy tuned to catch it would mostly fire on noise. With information-insensitive investors, price impact is pinned near $1/\xi$, intensities scale with ξ , and deviations become enormous relative to noise: in the Dou et al. [14] calibration ($\sigma_v = 1$, $\sigma_u = 0.1$, $\xi = 500$, $\theta = 0.1$, $I = 2$) the ratio is about 625. Detectability is therefore consistent with the regime split that Dou et al. [14] report (over-pruning only at $\xi = 0$, price triggers at $\xi = 500$), although, as we show below, detectability alone does not make Q-learners punish.

3.4 Can trigger strategies sustain collusion?

Detectability tells us whether a deviation can be seen; whether collusion can be an equilibrium depends also on what a deviation gains and what punishment costs. We ask this in the spirit of Green and Porter [21] and Abreu et al. [2]. On the collusive path every learner trades b per unit of v and the market maker prices at $\lambda(Ib)$. A deviator plays its static best response for one period. The market maker re-estimates price impact slowly, so its λ does not react within the period. Rivals observe last period's value and order flow and start a punishment phase when the order-flow surprise $\text{sign}(v)(y - Ibv)$ exceeds $k\sigma_u$. Noise triggers punishment by mistake with probability $q_0 = 1 - \Phi(k)$; a deviation at value v triggers it with probability $q_1(v) = 1 - \Phi(k - d(v))$, where $d(v) = (\beta^d - b)|v|/\sigma_u$.

Proposition 3 (Nash-reversion trigger strategies). *Let punishment be T periods of static Nash play ($T = \infty$: grim trigger), and let V be the value of the collusive regime including mistaken punishments,*

$$V = \frac{\pi(b) + \delta q_0 \pi^N w_T}{1 - \delta(1 - q_0) - \delta^{T+1} q_0}, \quad w_T = \frac{1 - \delta^T}{1 - \delta}.$$

Intensity b is sustainable with threshold k if and only if, at every value v ,

$$g(v) \leq \delta (q_1(v) - q_0) (1 - \delta^T) \left(V - \frac{\pi^N}{1 - \delta} \right),$$

where $g(v)$ is the one-period gain of the best-response deviation.

Proposition 4 (A bound for any punishment). *Whatever the strategies, a deviation at value v changes the distribution of everything rivals observe by at most the total-variation distance $2\Phi(d(v)/2) - 1$, and a trader's continuation value lies between 0 (it can always stop trading) and $I\pi^C/(1 - \delta)$ (with the market maker pricing rationally, no trader can earn more than the whole cartel). Hence a stationary outcome b can be sustained by some punishment scheme only if*

$$g(v) \leq \frac{\delta}{1 - \delta} (2\Phi(d(v)/2) - 1) I \pi^C \quad \text{for every } v.$$

Figure 2 evaluates both conditions, optimising the threshold k (proofs and computations in Appendix A). Three facts matter for what follows. First, in the standard Kyle market Nash reversion sustains *no* collusion for any $\delta \leq 0.99$, grim or not: a deviation is less than half a noise standard deviation, so detection is weak, while the collusive surplus $\pi^C - \pi^N$ is small relative to the deviation gain, and the mistaken punishments needed to catch deviations destroy the surplus faster than they deter (the Green and Porter [21] problem in its starkest form). Collusion is not impossible: the bound of Proposition 4 allows full collusion at $\delta = 0.95$, but only with punishments far harsher than reverting to Nash. Second, with many information-insensitive investors monitoring is almost perfect, and grim Nash reversion sustains full collusion once $\delta \geq 0.80$. Third, a one-period punishment, the most a strategy that remembers one period can impose by reverting to Nash, sustains at most $\Delta = 0.43$ at $\delta = 0.95$ (0.44 as $\delta \rightarrow 1$): the gain from deviating at the largest value exceeds one period of lost surplus.

These benchmarks sharpen the empirical question. In the standard market an outcome near the cartel can only be an equilibrium if traders punish much harder than Nash reversion, which the deviation test can see. In the Dou et al. [14] market, outcomes above 0.43 reached by learners that remember a single period cannot rest on Nash reversion either.

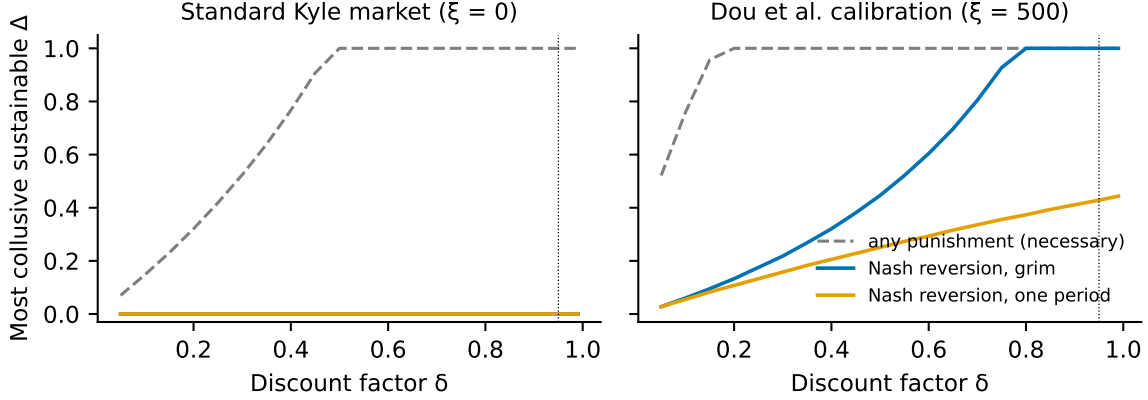


Figure 2: Most collusive outcome (in units of Δ intensity) that trigger strategies can sustain, by discount factor. Solid lines: static-Nash reversion with the best threshold, forever (grim) or for one period (the harshest phase a memory-one strategy can run). Dashed: the necessary condition of Proposition 4, which no scheme can beat. Dotted: $\delta = 0.95$, used in all experiments. Left: $\sigma_u = 1$, 5 values; right: Dou et al. [14]’s calibration, 10 values.

4 Learning Traders and Diagnostics

4.1 Learning algorithms

Tabular Q-learning. Each trader in each market owns a Q-table indexed by a memory state s (what it remembers about the previous period), the current value v , and an order from a finite grid. It plays ε -greedy with $\varepsilon_t = e^{-\beta t}$ and updates

$$Q(s_t, v_t, x_t) \leftarrow Q(s_t, v_t, x_t) + \alpha_t \left[\pi_t + \gamma \max_{x'} Q(s_{t+1}, v_{t+1}, x') - Q(s_t, v_t, x_t) \right].$$

Following Calvano et al. [7] and Dou et al. [14], Q is initialised to the discounted payoff of each order against uniformly random rivals, and the step size is constant ($\alpha_t = \alpha$) unless stated otherwise. As an alternative we use per-entry decaying step sizes $\alpha_n = \alpha(1 + n/\kappa)^{-0.7}$, where n counts updates of the entry, which satisfy the Robbins and Monro [27] conditions.

Deep RL. Double DQN [26, 32] and PPO with generalised advantage estimation [28, 29] use the same information as continuous features (current value, previous value, and the memory signal), the same order grid, and one independent network per trader per market.

Memory. We vary what traders remember from the previous period: *none* (no memory; punishment is impossible); *flow* (binned aggregate order flow); *residual* (previous value and binned flow net of own order, i.e. what everyone else and noise traders did); *orders* (previous value and the rival’s exact order: perfect monitoring); and *price* (previous value and binned price surprise, the state used by Dou et al. [14]).

Market maker. The market maker re-estimates λ_B from exponentially weighted moments of (v, y) with a half-life of 2,000 periods and prices with (1), so it responds to the learners’ strategies as they evolve.

4.2 Diagnostics

After training, exploration and learning are switched off and greedy play is evaluated for 20,000 periods while the market maker keeps adapting.

D1: rival-deviation test. From a point on the greedy path we clone each market. In the clone, trader 1 plays its myopic best response for one period; afterwards everyone follows the learned strategies in both copies, which share every value and noise draw. We report the rivals' change in trading intensity at each lag k , $\Delta\beta_k^{\text{rival}} = \mathbb{E}[v_k \Delta x_k^{\text{rival}}] / \text{Var}(v)$, and the deviator's discounted gain over the window. Punishment means $\Delta\beta_1^{\text{rival}} > 0$ and a negative gain. This is the test Calvano et al. [7] use for pricing algorithms, adapted to a stochastic market with paired simulations.

D2: noise-shock test. Following Dou et al. [14], the clone receives an extra noise-trader order in the direction of v , moving the price the way a rival's over-trading would although nobody deviated. Under price-trigger strategies traders trade harder next period; under over-pruning they do not react. We size shocks in *deviation units*: 1 is the flow change of a best-response deviation at $|v| = \mathbb{E}|v|$, so shocks mean the same thing in both regimes (Proposition 2).

D3: myopic placebo. We retrain the same learners with $\gamma = 0$. A myopic trader values only today's profit, so it gains nothing from punishing and fears no punishment. Its Q-values estimate immediate expected profit, and any systematic reaction it shows to deviations or shocks cannot be a punishment strategy. If myopic learners reproduce a "collusive" outcome or signature, that outcome or signature does not identify collusion. The placebo also has an exact benchmark:

Proposition 5 (Myopic rest points are competitive). *Let $\gamma = 0$ and take any rest point of learning at which each trader's greedy policy is fixed and its Q-values equal the expected one-period profit of each order given rivals' greedy play and the market maker's λ . Then at every memory state each greedy order is a best response on the grid to the rivals' orders, and the stage game (which does not depend on the state) has a unique equilibrium given λ , $\beta = 1/((n+1)\lambda)$. If the market maker prices rationally, play is the Nash benchmark up to grid resolution.*

Payoffs depend on the memory state only through what rivals do, and best responses $x_i = (v - \lambda \sum_{j \neq i} x_j) / (2\lambda)$ have slope $-1/2$, so the stage equilibrium is unique and the same in every state; the rational λ given β^N is λ^N . Consequently, whenever myopic learners settle away from Nash, their value estimates are biased, and the size of that departure measures the learning bias with no strategic component. The argument applies verbatim to any stage game with a unique equilibrium, including the logit Bertrand game of Calvano et al. [7] (Section 5.1).

D4: no-memory control. With memory *none*, traders cannot condition on the past, so no punishment strategy exists; any supracompetitive outcome is a learning bias.

D5: uninformative memory. We replace an informative memory by a *random* state of the same size, drawn independently each period from a separate random stream (so the value and noise draws are unchanged). If the outcome survives, the content of memory, which punishment would require, is not what produces it.

D6: a falsification test with passive traders. When passive Nash traders outnumber the learners by more than one, the cartel intensity lies *above* the Nash intensity (Remark 1). Collusion then predicts more trading and a learning bias predicts less, so the direction of the outcome discriminates between them.

D7: learning-rate sensitivity. A strategic equilibrium is a property of the game; a bias of the learning procedure should scale with the step size. We vary the constant α , compare with decaying step sizes, and replace the standard update with counterfactual updates, which remove the selection effect.

Convergence. Constant-step Q-learning with noisy payoffs keeps estimates moving, so greedy strategies need not settle. We report the share of greedy table entries that changed over the last 10^5 training periods and the mean on-path change in grid steps.

4.3 Experimental design and implementation

Unless noted, $I = 2$, $P = 0$, $\sigma_v = 1$, $\gamma = 0.95$, and each configuration is run for 100–200 independent markets. Standard-Kyle experiments ($\xi = 0$) use $\sigma_u = 1$, 5 values and a 31-point order grid spanning ± 1.5 times the largest Nash order; results in this regime are invariant to σ_u (Proposition 2). The Dou et al. [14] replication uses their calibration ($\sigma_u = 0.1$, $\theta = 0.1$, $\alpha = 0.05$), 10 values, a 15-point per-value grid spanning the cartel-to-Nash range plus 10%, and the price state. Confidence intervals are 95% across markets. The simulator advances all markets of a configuration as one batched array operation, which makes each experiment a matter of minutes to hours on a 4-core CPU. The code, configurations and a 72-test suite are released with the paper; the tests check the benchmarks against numerical optimisation, reproduce the Nash and cartel outcomes when the corresponding strategies are played, and verify that D1 and D2 report a reaction for scripted punishing traders and none for scripted memoryless ones.

5 Results

5.1 Do the diagnostics detect punishment where it exists?

A test that always says “no punishment” would be useless. Before applying the diagnostics to traders, we run them on the case where algorithmic collusion is held to be sustained by punishment: the logit Bertrand duopoly of Calvano et al. [7], with their baseline parameters (demand $a_i = 2$, $a_0 = 0$, $\mu = 1/4$, cost 1, 15 prices spanning the Nash-to-monopoly range plus 10% on each side, memory of both last prices, $\alpha = 0.15$, $\beta = 4 \times 10^{-6}$, $\delta = 0.95$; 100 sessions, 2.5×10^6 periods). The deviation test is the one used for traders: one firm plays its static best response to the rival’s greedy price for one period, paired with an undeviated copy. Table 1 reports the profit-based collusion index, the rival’s price change one period after the deviation (in units of the monopoly–Nash price gap), and the deviator’s discounted gain over 16 periods (in units of the per-period monopoly–Nash profit gap).

The baseline reproduces Calvano et al. [7]: $\Delta = 0.86$, and after a deviation the rival cuts its price (−0.54) and the deviator loses (−1.72). The test sees punishment when it is there. The placebos then behave as they should, and they also show why the price level is not enough. Myopic learners, who by Proposition 5 would play Nash if their estimates were unbiased, settle at $\Delta = 0.22$ and do not punish; the departure from Nash is their learning bias. Memoryless learners reach $\Delta = 0.95$, *higher* than the baseline, with no rival reaction at all and a profitable deviation (0.48): a price level

Table 1: Diagnostics outside the Kyle market. Logit Bertrand: the Calvano et al. [7] baseline and placebos. Dealers: two Q-learning market makers quoting asks against informed and liquidity buyers (Section 5.2). Rival response: price change one period after a forced best-response deviation, relative to the monopoly–Nash price gap. Deviator gain: discounted over 16 periods, relative to the per-period monopoly–Nash profit gap. Verdict “punishment” when the rival cuts its price and deviating loses, both significantly. 100 sessions; 95% CIs.

Learners	Δ profit	Δ best ask	Rival price response	Deviator gain	Verdict
<i>Logit Bertrand (Calvano et al.)</i>					
Baseline ($\gamma = 0.95$, memory)	0.86 ± 0.02	–	-0.54 ± 0.06	-1.72 ± 0.22	punishment
Myopic ($\gamma = 0$)	0.22 ± 0.02	–	-0.01 ± 0.05	0.01 ± 0.10	no punishment
No memory	0.95 ± 0.01	–	0.00 ± 0.00	0.48 ± 0.03	no punishment
Uninformative memory	0.053 ± 0.003	–	0.00 ± 0.00	0.07 ± 0.04	no punishment
Noisy profits ($\sigma = 0.1$)	0.79 ± 0.03	–	-0.46 ± 0.08	-1.96 ± 0.32	punishment
Noisy profits, myopic	0.03 ± 0.01	–	0.02 ± 0.04	0.07 ± 0.11	no punishment
Counterfactual updates	0.33 ± 0.02	–	-0.25 ± 0.03	-0.34 ± 0.05	punishment
<i>Dealers under adverse selection</i>					
Baseline ($\gamma = 0.95$, memory)	0.84 ± 0.02	0.64 ± 0.02	0.01 ± 0.06	1.07 ± 0.54	no punishment
Myopic ($\gamma = 0$)	0.974 ± 0.003	0.841 ± 0.004	-0.001 ± 0.001	0.89 ± 0.02	no punishment
No memory	0.86 ± 0.02	0.65 ± 0.03	0.00 ± 0.00	0.95 ± 0.12	no punishment
Uninformative memory	0.922 ± 0.002	0.742 ± 0.003	0.00 ± 0.00	1.05 ± 0.15	no punishment
Counterfactual updates	0.30 ± 0.03	0.19 ± 0.02	0.09 ± 0.09	0.58 ± 0.34	no punishment
Counterfactual, no memory	0.28 ± 0.06	0.17 ± 0.04	0.00 ± 0.00	0.36 ± 0.08	no punishment

indistinguishable from collusion with nothing behind it, the “spontaneous coupling” of Banchio and Mantegazza [5]. An uninformative memory of the same size as the real one collapses prices to $\Delta = 0.05$, so in this game what memory contains matters, as punishment requires. With noisy profits the baseline still gives $\Delta = 0.79$ and still punishes (myopic: 0.03). Counterfactual updates lower prices to $\Delta = 0.33$, as Asker et al. [3] find for synchronous learning, but what remains is still enforced: the rival cuts its price after a deviation (-0.25) and deviating loses (-0.34). The learning rule changes the level, not the nature, of collusion in this game.

The diagnostics therefore discriminate. They find punishment in the game where the literature finds it, and in the same game they flag a supracompetitive outcome without punishment when memory is removed. What they find among traders is not a property of the tests.

5.2 A second market: dealers under adverse selection

Q-learning market makers also sustain supracompetitive quotes [10]. We apply the same diagnostics to a stylised dealer market. Two dealers post ask quotes on a 15-point grid spanning the competitive and monopoly asks plus 10%; each period one buyer arrives. With probability $\pi = 0.3$ the buyer is informed and buys only when the asset is worth $v = +\sigma$ ($v = \pm\sigma$, $\sigma = 0.5$); otherwise it is a liquidity buyer with valuation $w \sim U[0, 1]$ who buys if w exceeds the best ask. The dealer with the lower ask serves the order (ties split) and earns ask minus v . The competitive ask is the zero-profit Glosten and Milgrom [20] quote ($a^N = 0.10$) and the monopoly ask maximises expected profit ($a^M = 0.61$). Feedback is noisy, as the buyer’s type and the asset value are random, and the profit every ask would have earned against the same buyer is known, so counterfactual updates apply exactly. Dealers remember both last quotes, as in Calvano et al. [7], and learn with the same parameters.

The lower panel of Table 1 reports the results. Dealers quote well above the competitive ask: the best ask closes 0.64 of the gap to the monopoly ask and dealers earn 0.84 of the monopoly profit gap. Nothing in this outcome is enforced. After a forced undercut the rival does not lower its

quote (0.01), and undercutting pays (+1.07). Every placebo reproduces the markup: myopic dealers quote even higher (0.84), as do dealers without memory (0.65) or with an uninformative one (0.74). Counterfactual updates, which remove the selection effect, bring the best ask most of the way to the competitive level (0.19; without memory 0.17). This is the pattern of a learning bias, and it agrees with the reading of Colliard et al. [10], who trace dealers’ markups to limited experimentation and noisy feedback rather than to punishment. The protocol separates the two markets cleanly: the same tests that find punishment in the pricing duopoly find none among dealers, although both reach similar price levels.

5.3 Pruning without a game

We start with the simplest possible setting: one informed trader ($I = 1$), a price rule frozen at the monopoly value, no memory, and $\gamma = 0$. There is no rival, no future, and nothing strategic; the optimal order is $v/(2\lambda)$. Figure 3 shows the learned intensity relative to the optimum across 100 independent runs per step size. Standard Q-learning under-trades, reaching only 0.87 of the optimal intensity at $\alpha = 0.02$ and 0.43 at $\alpha = 0.4$. The shortfall follows a square-root law, $1 - \hat{\beta}/\beta^* \approx 0.91\sqrt{\alpha}$ ($R^2 = 0.994$). Counterfactual updating, which credits every order with the profit it would have earned, learns the optimum at every step size (1.00–1.01).

The mechanism is the selection effect described in Section 2. With a constant step size, an action’s estimate is an exponentially weighted average of its recent payoffs and fluctuates with standard deviation proportional to $\sqrt{\alpha}$ times the payoff noise. In a Kyle market the payoff noise of an order is $\lambda\sigma_u|x|$, so larger orders have noisier estimates. An order whose estimate dips below the greedy order is updated only during exploration, so the dip persists; an order whose estimate rises is chosen and quickly corrected. The asymmetry acts like a penalty proportional to $\sqrt{\alpha}\lambda\sigma_u|x|$, which shifts the learned order toward zero by an amount proportional to $\sqrt{\alpha}$ (Appendix B gives a heuristic derivation). Counterfactual updates remove the selection, because every order is updated every period. What remains under counterfactual updates is the ordinary winner’s curse of acting on the largest of several noisy estimates, which favours the noisier, larger orders; consistent with this, counterfactual learners trade marginally more than the optimum here (1.00–1.01) and in the market ($\Delta = -0.10$, Section 5.6).

The bias also depends on the discount factor (Figure 3b). At $\alpha = 0.15$ the learned intensity rises from 0.64 of the optimum at $\gamma = 0$ to 0.73, 0.82 and 0.89 at $\gamma = 0.5, 0.8, 0.95$. With $\gamma > 0$ every update target also contains the continuation value $\gamma \max_{x'} Q(s', v', x')$, whose noise comes from the next value draw and is the same whatever order was submitted. This adds action-independent noise that dilutes the size-dependent payoff noise behind the selection effect. Myopic learners therefore prune more than forward-looking ones, for reasons that have nothing to do with strategy.

This is the over-pruning bias of Dou et al. [14], here isolated from any interaction: it alone produces trading intensities 11–57% below the optimum. Any test for collusion in a market of such learners must therefore separate this bias from strategic behaviour.

5.4 The standard Kyle market: low trading without punishment

Table 2 and Figure 4 report outcomes for two learners in the standard Kyle market ($\xi = 0$) with a constant step size $\alpha = 0.15$, 200 markets per seed and three seeds per condition.

Outcomes. Forward-looking learners that remember what their rivals and the noise traders did (*residual* memory) trade far below Nash: their aggregate intensity covers 0.86 of the distance from competition to the cartel (seed means 0.85–0.88), informativeness 0.93, and profit 0.46. With

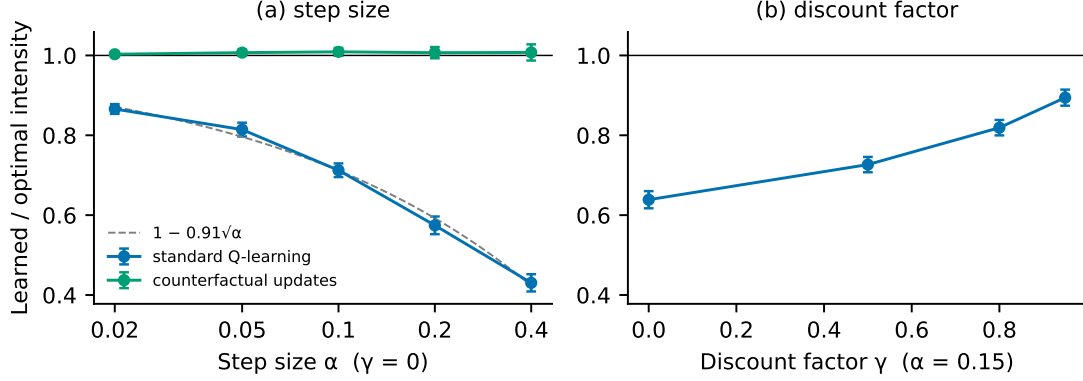


Figure 3: Pruning without a game. One informed trader facing a fixed price rule ($I = 1$, no memory, 3×10^5 periods, 100 runs per point, 95% CIs). (a) With $\gamma = 0$, standard Q-learning under-trades by about $0.91\sqrt{\alpha}$ (dashed); counterfactual updates learn the optimal order at every step size. (b) At $\alpha = 0.15$ the shortfall shrinks as the discount factor rises.

perfect monitoring of each other’s orders, the index is 0.82. Without memory, where punishment is impossible, learners still close 0.24 of the gap. Read conventionally, these numbers describe substantial collusion.

The myopic placebo. Retraining the residual-memory learners with $\gamma = 0$, which removes any reason to punish or to fear punishment, produces an intensity index of 1.24 and an informativeness index of 1.46: myopic learners trade *less than the cartel* and make prices less informative than full collusion would. They do not, however, earn collusive profits ($\Delta \text{ profit} = -0.01$), because trading below the cartel level gives up profit. A learner that cannot collude thus scores higher on the intensity- and informativeness-based indices than learners that might. These indices measure how little algorithms trade, not why, and cannot by themselves identify collusion.

Punishment tests. The rival-deviation test is the most direct evidence. Figure 5 shows that after a forced one-period best-response deviation, rivals’ trading intensity is unchanged at every lag in every condition: the reaction at the first lag is -0.000 ± 0.003 for residual memory and 0.008 ± 0.008 (about 3% of the deviation itself) with perfect monitoring. Deviating pays: the deviator’s discounted gain over the following 16 periods is 0.053 ± 0.003 with residual memory and 0.058 ± 0.007 with perfect monitoring, essentially all of it earned in the deviation period. No condition shows the punishment-then-reversion pattern that identifies collusion in Calvano et al. [7]. The outcome is supracompetitive but not self-enforcing, which is the signature of pruning.

This matches Proposition 2: at $\xi = 0$ a best-response deviation is about a quarter of a noise standard deviation per unit of v/σ_v , so a trigger on observed flow would fire mostly on noise. It is also consistent with the classification of Dou et al. [14], as reported in their reimplementation [15], which attributes collusion at $\xi = 0$ to over-pruning. What our diagnostics add is that even perfect monitoring, which removes the detectability problem entirely, does not lead Q-learners to discover punishment.

Convergence. With a constant step size and noisy payoffs, greedy strategies keep changing among neighbouring orders: on-path orders moved by 1.38 grid steps on average over the last 10^5 training

Condition	Δ intensity	Δ inform.	Δ profit	Rival $\Delta\beta_1$	$\Delta\beta_1/\Delta\beta_0^{\text{dev}}$	Deviator gain	Markets
Remembers rivals ($\gamma = 0.95$)	0.86 ± 0.01	0.93 ± 0.01	0.46 ± 0.02	-0.000 ± 0.003	-0.00	0.053 ± 0.003	600
Remembers rivals, myopic ($\gamma = 0$)	1.241 ± 0.005	1.46 ± 0.01	-0.01 ± 0.01	0.002 ± 0.003	0.00	0.124 ± 0.004	600
No memory ($\gamma = 0.95$)	0.24 ± 0.02	0.26 ± 0.02	0.16 ± 0.04	0.000 ± 0.000	0.00	0.018 ± 0.001	600
No memory, myopic ($\gamma = 0$)	1.27 ± 0.03	1.49 ± 0.03	-0.03 ± 0.05	0.000 ± 0.000	0.00	0.115 ± 0.003	600
Perfect monitoring ($\gamma = 0.95$)	0.82 ± 0.01	0.90 ± 0.01	0.39 ± 0.03	0.008 ± 0.008	0.03	0.058 ± 0.007	200

Table 2: Standard Kyle market ($\xi = 0$, $I = 2$, constant $\alpha = 0.15$). Δ : collusion indices (0 = Nash, 1 = cartel) pooled over markets and seeds. Rival $\Delta\beta_1$: change in rivals’ trading intensity one period after a forced best-response deviation; next column: the same as a fraction of the deviation. Deviator gain: discounted profit gain from deviating over 16 periods. 95% confidence intervals.

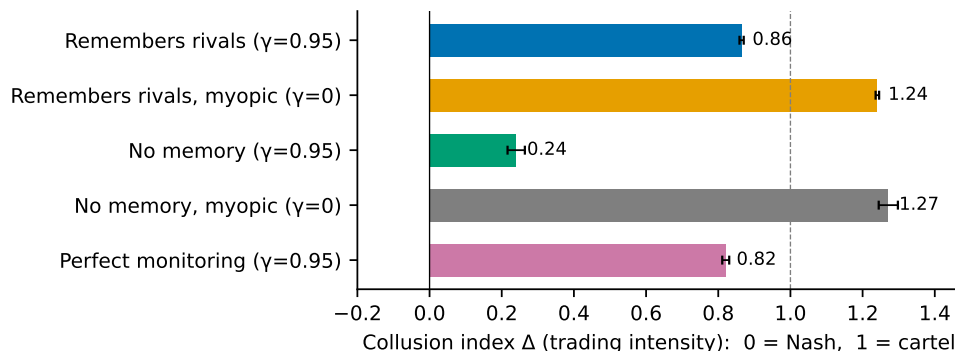


Figure 4: Collusion index (trading intensity) by condition. The myopic placebo, which cannot collude, scores beyond the cartel.

periods (residual memory). Outcomes are nevertheless stable: seed-to-seed variation in Δ is a few hundredths, and the residual-memory results are unchanged between 6×10^6 and 1.5×10^7 training periods (Appendix Table 9).

5.5 The Dou et al. regime: detectable deviations, still no punishment

We next replicate the Dou et al. [14] setting: their calibration ($\sigma_u = 0.1$, $\theta = 0.1$, $\alpha = 0.05$), 10 values, a 15-point order grid per value spanning the cartel-to-Nash range plus 10%, and the lagged price and value as state, at $\xi = 0$ and at $\xi = 500$, where Proposition 2 puts a best-response deviation at about 625 noise standard deviations. Each trader has its own Q-table, as in the paper’s description and the primary specification of Esquinas Coves [15]. At $\xi = 500$ the price state bins the lagged price over the range the order grid can produce, like the price grid of Dou et al. [14]; Appendix C shows that a state binned in units of noise saturates in this regime and reports both versions. Table 3 and Figure 6 report the results.

Levels. Forward-looking learners with price memory reach $\Delta = 0.47$ at $\xi = 0$ and 0.48 at $\xi = 500$. With the noise-unit state, the $\xi = 500$ value is 0.31, matching the 0.31 that Esquinas Coves [15] report for separate Q-tables in this regime. Our $\xi = 0$ value is below their 0.72; a likely reason, which we have not tested, is the training horizon: their runs last 5×10^8 periods, ours 1.5×10^7 .

No punishment, even when deviations are glaring. In both regimes the forward-looking learners ignore noise shocks: a shock as large as a full best-response deviation (515 noise standard deviations at $\xi = 500$) changes next-period intensity by $-0.00 \pm 0.49\%$ of β^N at $\xi = 500$, and forced

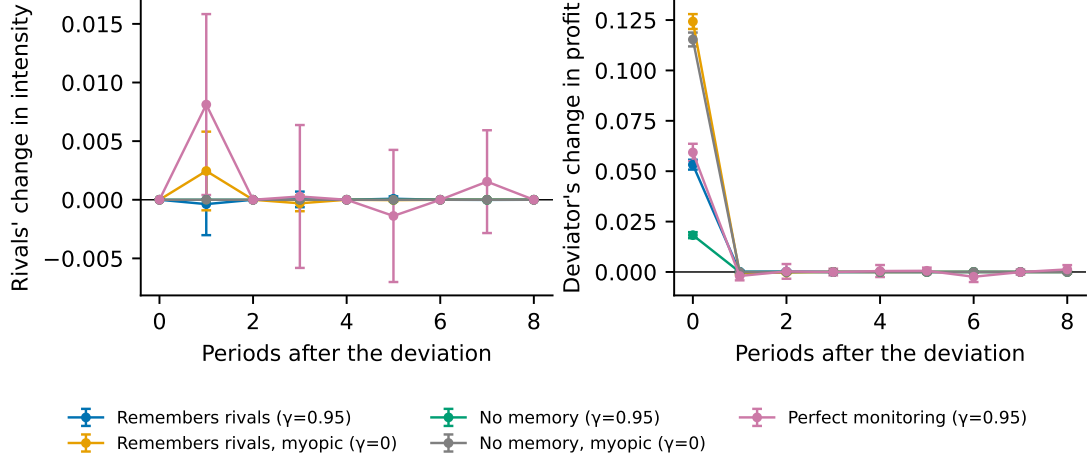


Figure 5: Rival-deviation test. Left: change in rivals’ trading intensity after trader 1 plays its one-period best response at lag 0. Right: the deviator’s change in profit. Paired simulations with common random numbers; 95% confidence intervals.

ξ	Separate Q-tables	Δ intensity	Δ profit	Shock response	Rival reaction	Seeds
0	$\gamma = 0.95$, remembers price	0.474 ± 0.002	0.59 ± 0.03	0.25 ± 0.39	-0.51 ± 0.37	2
0	$\gamma = 0$ (myopic placebo)	0.631 ± 0.002	0.75 ± 0.03	0.22 ± 0.39	0.10 ± 0.41	2
0	$\gamma = 0.95$, no memory	0.37 ± 0.02	0.50 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	2
500	$\gamma = 0.95$, remembers price	0.476 ± 0.002	0.64 ± 0.01	-0.00 ± 0.49	-0.48 ± 0.61	2
500	$\gamma = 0$ (myopic placebo)	0.130 ± 0.002	0.21 ± 0.01	0.69 ± 0.26	0.46 ± 0.33	2
500	$\gamma = 0.95$, no memory	0.23 ± 0.02	0.35 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	2

Table 3: Dou et al. [14] calibration with separate Q-tables (1.5×10^7 periods, 100 markets per seed). Shock response: change in per-trader intensity one period after a noise shock equal to one best-response deviation, in % of β^N . Rival reaction: change in rivals’ intensity one period after a forced best-response deviation, in % of β^N . At $\xi = 500$ the price state uses grid-range binning. The informativeness index is omitted because at $\xi = 500$ the Nash and cartel informativeness both equal one to four decimals. 95% CIs.

deviations draw no reaction from rivals ($-0.48 \pm 0.61\%$) and pay. Detectability is necessary for punishment but not sufficient: separate-table Q-learners do not discover trigger strategies within our horizon even when deviations are hundreds of standard deviations large.

The placebo. Myopic learners under-trade more than forward-looking ones at $\xi = 0$ (0.63 versus 0.47), as in the standard Kyle experiments, and less at $\xi = 500$ (0.13 versus 0.48). Neither shows a meaningful reaction to shocks or deviations, so these level differences are not about punishment.

Shared Q-tables and the placebo. Esquinas Coves [15] find that the price-trigger signature at $\xi = 500$ is sharpest when both traders read and update a single *shared* Q-table, and that this protocol also yields much higher collusion indices (0.87–0.92). A shared table is one learner choosing for several traders, so we apply the same placebo to it. Table 4 and Figure 7 show the results. Sharing raises the index for forward-looking learners to 0.68 ($\xi = 0$) and 0.62 ($\xi = 500$), and for myopic learners to 0.86 and 0.71.

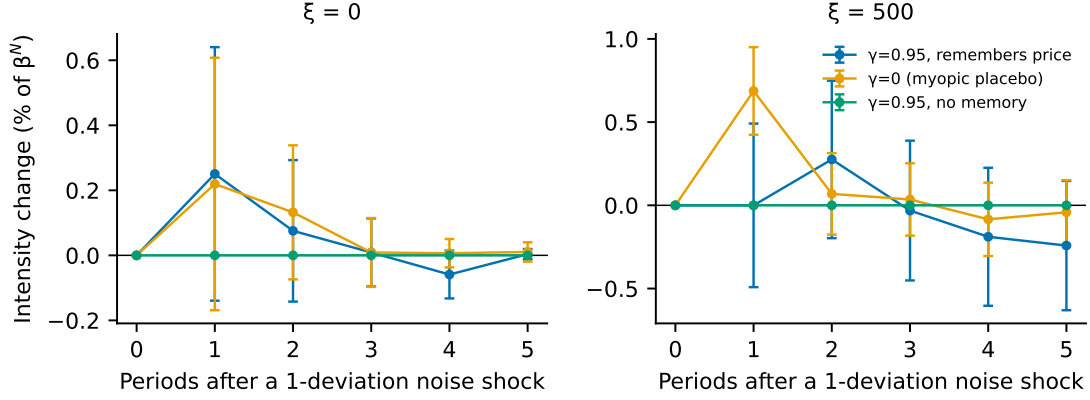


Figure 6: Noise-shock impulse responses in the Dou et al. [14] calibration with separate Q-tables. A noise-trader shock equal to one best-response deviation hits at lag 0; lines show the change in per-trader intensity in % of β^N . Under price-trigger strategies intensity would jump at lag 1.

ξ	Shared Q-table	Δ intensity	Shock response (% of β^N)			Rival reaction (% of β^N)	Seeds
			0.05	0.25	1		
0	$\gamma = 0.95$	0.683 ± 0.002	0.10 ± 0.11	-0.02 ± 0.24	0.12 ± 0.47	0.06 ± 0.36	2
0	$\gamma = 0$ (myopic placebo)	0.856 ± 0.001	0.02 ± 0.10	0.12 ± 0.16	0.46 ± 0.38	0.06 ± 0.31	2
500	$\gamma = 0.95$	0.618 ± 0.002	0.11 ± 0.40	0.11 ± 0.53	-0.30 ± 0.51	0.36 ± 0.55	2
500	$\gamma = 0$ (myopic placebo)	0.708 ± 0.001	0.30 ± 0.10	1.04 ± 0.16	4.01 ± 0.22	2.93 ± 0.17	2

Table 4: Shared Q-tables in the Dou et al. [14] calibration (1.5×10^7 periods, 100 markets per seed; grid-range price state at $\xi = 500$). Shock response: change in per-trader intensity one period after a noise shock of the given size in deviation units. Rival reaction: change in rivals' intensity one period after a forced best-response deviation. Both in % of β^N ; 95% CIs.

The punishment tests then fail the placebo. At $\xi = 500$ the *myopic* shared learner reacts to noise shocks in proportion to their size (0.30, 1.04 and 4.01% of β^N for shocks of 0.05, 0.25 and one deviation unit), the threshold-and-growth pattern that Dou et al. [14] read as a price trigger, and rivals trade harder after a forced deviation ($2.93 \pm 0.17\%$ of β^N), the pattern that Calvano et al. [7] read as punishment. A learner with $\gamma = 0$ cannot be punishing anyone, and indeed deviating still pays. The forward-looking shared learner shows neither reaction at our horizon. The responses come from the states the shocks and deviations lead to: they move the market into price states that greedy play visits less often, which have been pruned less and in which the learner therefore trades more aggressively; never-visited states keep orders close to their Nash-like initialisation (Appendix D). Reactions to shocks or deviations therefore identify punishment only if a myopic learner trained in the same way does not show them.

5.6 What changes the outcome: information or learning design?

The two channels make opposite predictions about interventions. If low trading were sustained by punishment, degrading the information traders use to monitor each other should restore competition, and the content of their memory should matter. If it is pruning, information is irrelevant and changes to the learning procedure should matter. Table 5 and Figure 9 report treatments in the standard Kyle market, each against the residual-memory baseline ($\Delta = 0.86$).

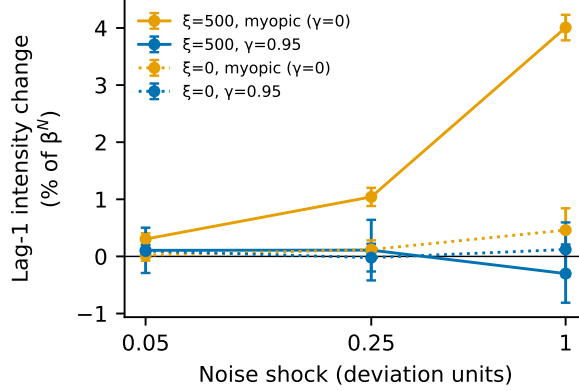


Figure 7: Noise-shock responses with shared Q-tables. At $\xi = 500$ the myopic placebo (orange, solid) ignores small shocks and reacts increasingly to larger ones, the pattern read as a price trigger, although with $\gamma = 0$ it has no reason to punish.

Information does not matter. Blurring the order flow that traders observe with noise of twice the noise-trading volume leaves the outcome unchanged ($\Delta = 0.89$). Replacing the informative memory with a *random* state that carries no information at all, but has the same number of states, reproduces it ($\Delta = 0.84$). The difference between forward-looking learners with and without memory (0.86 versus 0.24) is therefore a matter of table size: more states mean fewer updates per entry and more room for the selection effect of Section 5.3, not more information to coordinate on.

Learning design matters. The step size alone sets the level of the outcome (Figure 8): with residual memory the index is 0.19, 0.86 and 1.12 at $\alpha = 0.05, 0.15$ and 0.3 , and without memory $-0.03, 0.24$ and 0.51 , running from competition to beyond the cartel, with no punishment at any step size. Learning parameters can also change which strategic equilibrium a learner reaches, but a monotone path from competition to beyond the cartel, with no punishment at any point, is what the square-root law of Section 5.3 predicts. Two further changes to the learning rule move the outcome substantially. Counterfactual updating, which removes the selection effect, eliminates the supracompetitive outcome: forward-looking learners end at $\Delta = -0.10$ with residual memory and -0.09 without, that is, at (marginally beyond) the competitive level, with price informativeness at its Nash value (Δ informativeness -0.01). The market, the agents’ information and the exploration schedule are unchanged; only the update rule differs. Per-entry decaying step sizes change even the sign: with the schedule of Section 4, learners *over-trade* relative to Nash ($\Delta = -0.50$ with residual memory, -0.60 without), because step sizes shrink while the market maker’s price impact is still depressed by heavy exploration, freezing estimates around best responses to a market that is too cheap to trade in. These learners are not converged either (on-path orders still move by 4.74 grid steps), which underlines how strongly the learning procedure, rather than the game, shapes the outcome.

When collusion means trading more. Adding two passive informed traders who play the Nash strategy reverses the benchmarks: facing outside competition, the two learners would maximise their joint profit by trading *more* than their Nash intensity (aggregate 1.41 versus 1.00), because extra informed flow lowers the market maker’s price impact. A collusive force should push them up. Instead they trade 0.48, half their Nash intensity, moving away from the cartel and earning 30%

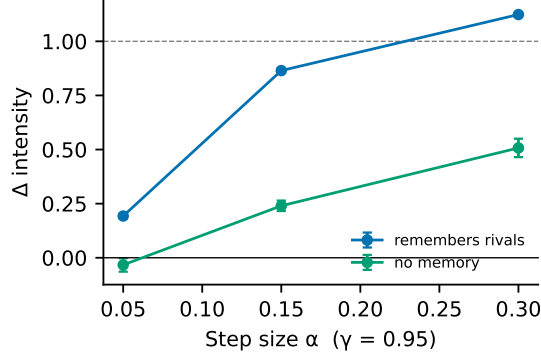


Figure 8: The collusion index rises with the constant step size α (standard Kyle market, $\gamma = 0.95$, 200 markets per point). No configuration punishes deviations.

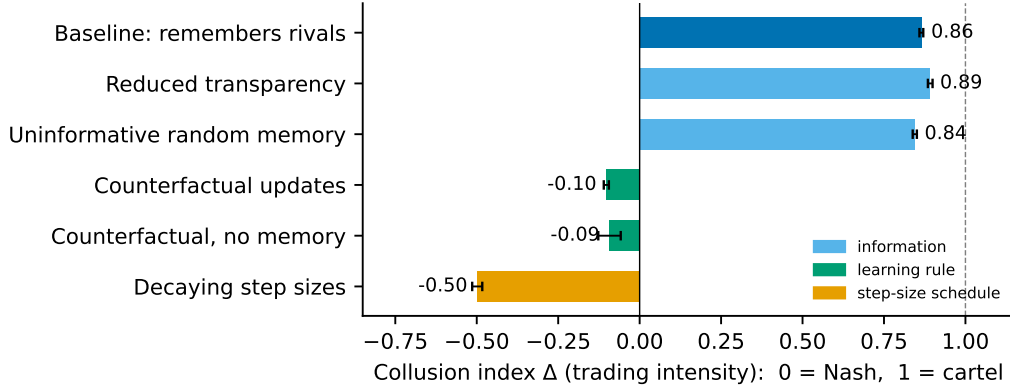


Figure 9: Interventions in the standard Kyle market. Treatments that change what traders observe leave the outcome unchanged; treatments that change how they learn move it to, or past, competition.

less than in the Nash equilibrium. The learners under-trade whichever side of Nash the collusive outcome lies on, which is what a learning bias predicts and the opposite of what collusion predicts. (In the collusion index this appears as $\Delta = -1.26$, which is why Table 5 should be read with the benchmarks in mind.)

5.7 Deep reinforcement learning

Table 6 repeats the standard-Kyle experiment with residual memory for Double DQN and PPO (4×10^5 periods, 50 markets each; the deep agents are far more expensive per step). Neither produces a supracompetitive outcome: PPO ends at the competitive level ($\Delta = -0.02$) and DQN slightly *over-trades* ($\Delta = -0.36$), with myopic variants close by (-0.28 and -0.17). No variant punishes deviations. This is what the mechanism of Section 5.3 predicts. Pruning requires estimates that stay frozen for orders that are rarely chosen. PPO learns from every sampled order through the policy gradient, with an entropy bonus that keeps sampling broad, and DQN’s network generalises across neighbouring orders and reuses past experience through replay, so neither leaves individual orders stranded with pessimistic estimates. We conjecture that DQN’s residual over-estimation from the max operator [30, 32], which favours orders with noisier payoffs, explains its slight over-trading. The

Treatment ($\xi = 0, I = 2$)	Δ intensity	Δ profit	Rival $\Delta\beta_1$	Deviator gain
Random memory, 35 states	0.84 ± 0.01	0.45 ± 0.02	0.000 ± 0.000	0.052 ± 0.003
Random memory, myopic	1.16 ± 0.01	0.11 ± 0.02	0.000 ± 0.000	0.107 ± 0.006
Counterfactual updates, residual memory	-0.10 ± 0.01	-0.44 ± 0.03	-0.001 ± 0.002	0.020 ± 0.003
Counterfactual updates, no memory	-0.09 ± 0.03	-0.34 ± 0.08	0.000 ± 0.000	0.015 ± 0.002
Counterfactual updates, residual, myopic	-0.10 ± 0.01	-0.44 ± 0.03	-0.001 ± 0.002	0.019 ± 0.003
Reduced transparency (disclosure noise $2\sigma_u$)	0.89 ± 0.01	0.45 ± 0.02	-0.001 ± 0.002	0.058 ± 0.003
Two passive Nash traders	-1.26 ± 0.01	-8.56 ± 0.09	0.001 ± 0.003	0.036 ± 0.003
DQN	-0.36 ± 0.07	-0.89 ± 0.13	-0.000 ± 0.004	0.026 ± 0.008
DQN, myopic	-0.17 ± 0.03	-0.46 ± 0.06	-0.001 ± 0.005	0.014 ± 0.007
PPO	-0.02 ± 0.09	-0.27 ± 0.12	0.000 ± 0.000	0.028 ± 0.008
PPO, myopic	-0.28 ± 0.05	-0.51 ± 0.10	0.000 ± 0.000	0.012 ± 0.006

Table 5: Treatments in the standard Kyle market ($\xi = 0, I = 2$, residual memory unless noted, 6×10^6 periods, 200 markets). Baseline: Δ intensity 0.86. 95% CIs.

Learner (residual memory)	Δ intensity	Δ informativeness	Rival $\Delta\beta_1$	Deviator gain
Tabular Q, $\gamma = 0.95$	0.86 ± 0.01	0.93 ± 0.01	-0.000 ± 0.003	0.053 ± 0.003
Tabular Q, $\gamma = 0$	1.241 ± 0.005	1.46 ± 0.01	0.002 ± 0.003	0.124 ± 0.004
DQN, $\gamma = 0.95$	-0.36 ± 0.07	-0.17 ± 0.05	-0.000 ± 0.004	0.026 ± 0.008
DQN, $\gamma = 0$	-0.17 ± 0.03	-0.09 ± 0.02	-0.001 ± 0.005	0.014 ± 0.007
PPO, $\gamma = 0.95$	-0.02 ± 0.09	0.06 ± 0.07	0.000 ± 0.000	0.028 ± 0.008
PPO, $\gamma = 0$	-0.28 ± 0.05	-0.18 ± 0.04	0.000 ± 0.000	0.012 ± 0.006

Table 6: Learning algorithms in the standard Kyle market ($\xi = 0, I = 2$, residual memory). Tabular Q: 600 markets, 6×10^6 periods; DQN and PPO: 50 markets, 4×10^5 periods. 95% CIs.

deep-RL horizon is short and the hyperparameters standard, so we view these results as evidence that the supracompetitive outcome is specific to tabular, constant-step value learning, not as a definitive statement about deep RL in markets [cf. 13].

6 An Audit Protocol

The diagnostics can be assembled into a protocol for deciding whether a supracompetitive outcome produced by learning algorithms is collusion. It needs what a firm testing its own algorithm, or a supervisor with access to a testing environment, typically has: the ability to retrain the algorithm and to replay a market from a saved state. Investment firms in the European Union must already test trading algorithms in such environments before deployment [16], so the protocol adds questions, not infrastructure.

Step 1: Benchmark. Compute the competitive and joint-profit benchmarks of the environment and the collusion index Δ . Only an outcome with Δ clearly above zero needs the remaining steps.

Step 2: Feasibility. Ask whether punishment could sustain the outcome at all (Propositions 3 and 4). An outcome above what any punishment can sustain is not equilibrium collusion, whatever else is true; one above what Nash reversion can sustain requires harsher punishments, which Step 3 would see.

Step 3: Deviation test. Force a one-period best-response deviation in a replayed market and

Table 7: The audit protocol applied to each market. Δ : profit-based (Bertrand, dealers) or intensity-based (Kyle). Deviation: does the rival respond and does deviating lose? Placebo: does a learner that cannot punish reproduce the outcome? Rule: does the outcome change with the learning rule (counterfactual updates or step size)?

Market	Learners	Δ	Rival responds	Deviating	Placebo Δ (no punishment possible)	Counterfactual Δ	Verdict
Logit Bertrand	memory one	0.86	yes (−0.54)	loses (−1.72)	myopic 0.22, no memory 0.95	0.33	collusion
Logit Bertrand	no memory	0.95	no	pays (+0.48)	(is a placebo)	–	learning bias
Kyle, $\xi = 0$	residual memory	0.86	no	pays (+0.053)	myopic 1.24, random memory 0.84	−0.10	learning bias
Kyle, $\xi = 0$	perfect monitoring	0.82	no	pays (+0.058)	as above	–	learning bias
Kyle, $\xi = 500$	separate tables	0.48	no	pays	myopic 0.13	–	learning bias
Kyle, $\xi = 500$	shared table	0.62	no	pays	myopic 0.71 (reacts to shocks)	–	learning bias
Dealers	memory one	0.84	no	pays (+1.07)	myopic 0.97, no memory 0.86	0.30	learning bias

compare it with an undeviated copy under common random numbers. Collusion predicts that rivals respond and that deviating does not pay over the following periods.

Step 4: Placebos. Retrain (a) myopically ($\gamma = 0$), (b) without memory and (c) with an uninformative memory of the same size. None of these can sustain punishment, and by Proposition 5 (a) plays Nash if its estimates are unbiased. Collusion predicts that the outcome, and any reaction to deviations or shocks, disappears.

Step 5: Learning rule. Retrain with a different step size and with counterfactual updates. An equilibrium is a property of the game and should not scale with the step size; a learning bias should.

The verdict is *collusion* if rivals respond to the deviation and deviating does not pay (Step 3), and the learners that cannot punish (Step 4) show no such response; *learning bias* if deviating pays and the outcome is reproduced by a placebo or moves with the learning rule (Steps 4 and 5); and inconclusive otherwise. The verdict rests on the response to deviations, not on the level of prices or trading: Steps 4 and 5 show how easily that level is produced without punishment. Each step can fail, and a genuine punishment scheme passes all of them.

Table 7 applies the protocol to every market in this paper. It returns *collusion* in exactly one case: the Calvano et al. [7] baseline. Every case among informed traders, the dealers, and the memoryless pricing duopoly return *learning bias*, although several of them would pass a test based on the price or trading level alone, and one (the myopic shared-table learner at $\xi = 500$) would pass a test based on the reaction to price shocks.

7 Discussion

What the diagnostics show. Four findings recur across our experiments. First, the supracompetitive outcomes of Q-learning informed traders do not rest on punishment: rivals do not react to deviations, even with perfect monitoring or when deviations are hundreds of noise standard deviations large, and deviating pays. Second, the same learning procedure under-trades with no rival at all, by an amount that grows with $\sqrt{\alpha}$ and disappears under counterfactual updates; in the market, counterfactual updating restores competition while degrading information does nothing. Third, placebos that cannot collude reproduce the conventional evidence of collusion: myopic learners score at or beyond the cartel on intensity- and informativeness-based indices, an uninformative memory reproduces the effect of an informative one, and a myopic shared-table learner shows the threshold-shaped reaction to price shocks that has been read as a price trigger. Fourth, detectability

rather than learning sophistication decides whether punishment is even feasible: in the standard Kyle market a deviation is a fraction of a noise standard deviation (Proposition 2).

Implications for auditing algorithms. Section 6 turns the diagnostics into a protocol. Its value is that every step can fail: applied to the Calvano et al. [7] baseline it finds punishment, and applied to informed traders it does not.

Implications for policy. If supracompetitive trading is a learning bias, interventions aimed at collusion through information, such as delayed or coarser order-flow disclosure, will not work; our reduced-transparency treatment confirms this. What works is algorithm design: counterfactual (full-information) updating removes the bias. This mirrors conclusions of Asker et al. [3, 4] and Abada and Lambin [1] for pricing algorithms and extends them to informed trading. A bias is not harmless: it still lowers liquidity and price informativeness, myopic learners push trading below even the cartel level, and with passive traders it costs the learners themselves 30% of their equilibrium profit. But its remedy is a design standard, not a cartel investigation.

Limitations. Our evidence concerns tabular Q-learning with standard hyperparameters and, more briefly, DQN and PPO; learners that model their rivals may find punishment strategies these do not. Constant-step Q-learning does not converge in the strict sense of Calvano et al. [7]: greedy strategies still change on path by one to two grid steps over the last 10^5 periods, although outcomes are stable across seeds, schedules and horizons. Our training horizons (6×10^6 to 1.5×10^7 periods) are shorter than those of Dou et al. [14] and Esquinas Coves [15] (up to 5×10^{10} and 5×10^8), and the forward-looking shared-table learner, which in their runs shows a price-trigger signature at $\xi = 500$, shows none at our horizon; longer training could change this comparison, which is why the placebo, not the absence of a response, carries our argument there. The market maker learns its price impact from exponentially weighted moments rather than being fixed at the equilibrium value. Finally, the stylised market abstracts from inventory dynamics, multiple assets and long-lived information.

Conclusion. Low trading by learning algorithms is easy to produce and easy to misread. Telling collusion from learning bias requires diagnostics that can fail: a deviation test, placebos that cannot collude, and a check of whether the outcome moves with the learning rule. In the canonical market for informed trading, and in the regime where deviations are most visible, these diagnostics point to pruning, not punishment.

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A Proofs

Proposition 1. With $x_j = \beta_j v$ and $p = \lambda y$, trader i ’s expected profit is $\beta_i \sigma_v^2 (1 - \lambda \sum_j \beta_j)$ (the noise term has zero mean). Taking λ as given, the first-order condition gives $\beta_i = (1 - \lambda \sum_{j \neq i} \beta_j) / (2\lambda)$, so a symmetric equilibrium has $\beta = 1 / ((n + 1)\lambda)$. With $\xi = 0$, rational pricing gives $\lambda = n\beta\sigma_v^2 / (n^2\beta^2\sigma_v^2 + \sigma_u^2)$. Substituting and solving yields $\beta^N = \sigma_u / (\sqrt{n}\sigma_v)$ and $\lambda^N = \sqrt{n}\sigma_v / ((n + 1)\sigma_u)$. For the collusive benchmark, total learner profit with rational pricing and aggregate $T = B_L + B_P$ is $B_L\sigma_v^2\sigma_u^2 / (T^2\sigma_v^2 + \sigma_u^2)$. The first-order condition in B_L is $T^2\sigma_v^2 + \sigma_u^2 - 2B_L T\sigma_v^2 = 0$, i.e. $\sigma_v^2(B_P^2 - B_L^2) + \sigma_u^2 = 0$, which gives B_L^C . For $\xi > 0$, $h(\beta) = (n + 1)\beta\lambda(n\beta) - 1$ is strictly increasing because both $\beta\lambda_B(n\beta) = n\beta^2\sigma_v^2 / (n^2\beta^2\sigma_v^2 + \sigma_u^2)$ and $\xi\beta$ are increasing, and $h(0) = -1$, $h(\infty) > 0$, so the root is unique. $\square$

Proposition 2. (i) With $\xi = 0$ and $P = 0$, $\beta^C = \sigma_u/(I\sigma_v)$ and $\lambda^C = \sigma_v/(2\sigma_u)$. Then $\beta^d = (1 - \lambda^C(I-1)\beta^C)/(2\lambda^C) = \frac{\sigma_u}{\sigma_v}(1 - \frac{I-1}{2I}) = \frac{\sigma_u}{\sigma_v} \cdot \frac{I+1}{2I}$, so $\beta^d - \beta^C = \frac{\sigma_u}{\sigma_v} \cdot \frac{I-1}{2I}$ and $\text{SNR} = (I-1)/(2I)$. (ii) From (1), $\lambda = 1/\xi + O(\xi^{-2})$ uniformly in $\lambda_B \in [0, \sigma_v/(2\sigma_u)]$. Treating λ as constant, the collusive aggregate maximises $B(1 - \lambda B)$, so $\beta^C = 1/(2I\lambda)$, and $\beta^d = (1 - \lambda(I-1)\beta^C)/(2\lambda) = \frac{1}{2\lambda} \cdot \frac{I+1}{2I}$. Hence $\beta^d - \beta^C = \frac{1}{\lambda} \cdot \frac{I-1}{4I} = \xi \frac{I-1}{4I}(1 + o(1))$ and $\text{SNR} = \xi \sigma_v(I-1)/(4I\sigma_u)(1 + o(1))$. Numerically, the exact ratio at $I = 2$, $\sigma_v = 1$, $\sigma_u = 0.1$, $\theta = 0.1$ is 62.49, 625.00 and 6250.0 for $\xi = 50, 500, 5000$. $\square$

Proposition 3. In the collusive regime a trader earns $\pi(b)$ today; with probability $1 - q_0$ the regime continues and with probability q_0 a punishment phase of T periods at π^N starts, after which the regime resumes. Hence $V = \pi(b) + \delta(1 - q_0)V + \delta q_0(\pi^N w_T + \delta^T V)$, which gives the expression for V . A one-period deviation at value v earns $\pi(b) + g(v)$ today, with $\pi(b)$ replaced by its value- v counterpart, and triggers punishment with probability $q_1(v)$. It is unprofitable iff $g(v) \leq \delta(q_1 - q_0)[V - (\pi^N w_T + \delta^T V)] = \delta(q_1 - q_0)(1 - \delta^T)(V - \pi^N/(1 - \delta))$. By the one-shot deviation principle it suffices to check one-period deviations from the collusive regime; during punishment, static Nash play is a stage equilibrium, so no deviation pays there. Profits are evaluated with λ fixed at its on-path value during the deviation period, since the market maker's estimate moves by a fraction $1 - 2^{-1/2000}$ per period, and with the rational λ^N during punishment. In Figure 2, b is scanned on 400 points from β^C to β^N , k on a grid from -3 to 6 in steps of 0.05 , and v ranges over the value grid used in the experiments. $\square$

Proposition 4. Fix any strategy profile and a history at which the path prescribes b , and let $W(\omega)$ be the deviator's continuation value as a function of everything observed at the end of the period, ω . Under conformity and deviation ω has distributions P_0 and P_1 that differ only through aggregate order flow, which is Gaussian with the same variance σ_u^2 and means that differ by $(\beta^d - b)v$. For any W with values in $[W_{\min}, W_{\max}]$, $\mathbb{E}_{P_0} W - \mathbb{E}_{P_1} W \leq \text{TV}(P_0, P_1)(W_{\max} - W_{\min})$, and $\text{TV} = 2\Phi(d(v)/2) - 1$ for two Gaussians whose means differ by d standard deviations. Since a trader can guarantee zero by not trading, $W_{\min} \geq 0$; since with rational pricing the learners' joint profit never exceeds $I\pi^C$ per period, $W_{\max} \leq I\pi^C/(1 - \delta)$. The deviation is unprofitable only if $g(v) \leq \delta \text{TV}(W_{\max} - W_{\min})$. $\square$

B Why pruning scales with $\sqrt{\alpha}$: a heuristic

Consider one trader, value $v > 0$, a fixed price rule with impact λ , and orders x with expected profit $\mu(x) = vx - \lambda x^2$ and payoff noise $\sigma(x) = \lambda \sigma_u x$. With a constant step size, the estimate of an order updated at every visit is an exponentially weighted average of past payoffs, with stationary standard deviation $s(x) = \sigma(x)\sqrt{\alpha/(2 - \alpha)} \approx \sigma(x)\sqrt{\alpha/2}$.

Under ε -greedy play the greedy order is updated almost every period, so its estimate stays close to μ . A competing order is updated only when explored, so after an update its estimate stays frozen for a long time. If the frozen estimate is below the greedy one the order stays non-greedy until the next exploration; if it is above, the order becomes greedy, is updated repeatedly, and its estimate is pulled back to μ . The greedy order is therefore, roughly, the one that keeps winning these comparisons, which favours orders whose estimates are *reliably* high: to first order the learner behaves as if it maximised $\mu(x) - c s(x)$ for a constant $c > 0$ that depends on the exploration schedule. With $s(x) \propto \sqrt{\alpha} \lambda \sigma_u x$, the maximiser is

$$x_\alpha^* = \frac{v - c' \sqrt{\alpha} \lambda \sigma_u}{2\lambda}, \quad 1 - \frac{x_\alpha^*}{x_0^*} = c' \sqrt{\alpha} \frac{\lambda \sigma_u}{v},$$

a shortfall linear in $\sqrt{\alpha}$, as found in Figure 3a ($R^2 = 0.994$). Counterfactual updates refresh every order every period, so no order is ever frozen and the selection disappears. With $\gamma > 0$, each target also carries continuation-value noise that does not depend on x ; this raises $s(x)$ equally for all orders and makes the comparison less sensitive to the size-dependent part, which is why the shortfall shrinks with γ (Figure 3b). We do not offer this as a proof; it is the intuition behind the selection effect, and the experiments test its predictions. Two pieces of it do rest on standard results. With a constant step size and persistent exploration, the mean dynamics of Q-learning in a stationary single-agent problem have the true expected payoffs as their unique rest point, and constant-step stochastic approximation keeps the iterates within a distance of order $\sqrt{\alpha}$ of the rest point in distribution [6, 24]. A departure of the greedy order from the optimum is therefore at most of order $\sqrt{\alpha}$, the scaling we find, and vanishes as $\alpha \rightarrow 0$. What these results do not give is the *sign* of the departure, which is what the selection argument supplies and what Figure 3a measures.

C Price-state binning at $\xi = 500$

Our first replication binned the lagged price as a surprise relative to the Nash-expected price, in units of noise-driven price variation (± 2.5 standard deviations, 15 bins). At $\xi = 500$ this saturates: learners that trade below Nash move the price hundreds of noise standard deviations from the Nash-expected value, so greedy play visits only the two outer bins. The main text therefore uses a grid-range state like the price grid of Dou et al. [14]: 15 equal bins over the range of prices the order grid can produce given v_{t-1} , with a $\pm 2\sigma_u$ margin. Table 8 reports both. The qualitative conclusions are the same: forward-looking learners never react to shocks or deviations, and the myopic shared learner reacts to large shocks under both binnings.

Learners ($\xi = 500$)	Price state	Δ intensity	Shock response (1 dev.)	Rival reaction	Seeds
Separate, $\gamma = 0.95$	noise units	0.31 ± 0.01	-0.59 ± 0.51	-0.17 ± 0.38	1
Separate, $\gamma = 0.95$	grid range	0.476 ± 0.002	-0.00 ± 0.49	-0.48 ± 0.61	2
Separate, $\gamma = 0$	noise units	0.113 ± 0.004	-0.02 ± 0.32	-0.34 ± 0.38	1
Separate, $\gamma = 0$	grid range	0.130 ± 0.002	0.69 ± 0.26	0.46 ± 0.33	2
Shared, $\gamma = 0.95$	noise units	0.71 ± 0.01	0.24 ± 0.50	0.08 ± 0.20	1
Shared, $\gamma = 0.95$	grid range	0.618 ± 0.002	-0.30 ± 0.51	0.36 ± 0.55	2
Shared, $\gamma = 0$	noise units	0.863 ± 0.003	3.34 ± 0.32	0.15 ± 0.09	1
Shared, $\gamma = 0$	grid range	0.708 ± 0.001	4.01 ± 0.22	2.93 ± 0.17	2

Table 8: $\xi = 500$ results under the two price-state binnings. Shock response and rival reaction in % of β^N ; 95% CIs.

D Rare states and the shock response

To see why a myopic shared-table learner reacts to large noise shocks and to deviations at $\xi = 500$, we trained it once more under each price-state binning (20 markets, 1.5×10^7 periods, otherwise as in Table 4), recorded which states greedy play visits, and computed the greedy trading intensity in every state on a scale where 0 is the cartel and 1 is Nash.

Noise-unit binning. Greedy play visits only the two outermost price-surprise bins (100% of visits), because learners that trade below Nash move the price hundreds of noise standard deviations

from the Nash-expected value. In those states the learned intensity is close to the cartel (0.29–0.29). The thirteen interior bins are never visited, and their greedy orders remain near their Nash-like initialisation (0.79–0.81). A shock in the direction of v moves the next state towards the interior; small shocks do not reach it, large ones do, which produces the threshold.

Grid-range binning. Greedy play now spreads over all fifteen price bins (3.8–9.8% of visits each), and the gradient is milder but has the same sign: intensity is 0.30 in states visited more often than the median, 0.35 in states visited less often, and 0.47 in (value, price) states never visited in greedy play. Learned intensity also varies across price bins (0.31–0.42) for reasons unrelated to rivals’ behaviour, since with $\gamma = 0$ each state’s Q-values only estimate immediate profit.

Interpretation. With pruning, how aggressively a Q-learner trades in a state depends on how often, and how recently, that state has been updated: heavily visited states are pruned most, rarely visited ones less, and never-visited ones not at all. A shock or a rival’s deviation moves the market into a different price state, and the learner’s trading changes with it. The change looks like a reaction to the shock or the deviation, but it reflects the uneven training of the states, not a strategy, which is why it appears in a learner with $\gamma = 0$. The general lesson does not depend on the binning: whenever shocks or deviations move a Q-learner into less-trained states, its reaction reflects initialisation and incomplete exploration, and only a placebo can tell it apart from punishment.

E Robustness

Table 9 reports the collusion indices of forward-looking Q-learners in the standard Kyle market across training horizons, exploration schedules and seeds. Outcomes are stable once exploration is long enough; the residual-memory run with the fastest exploration decay is the exception because its larger table is under-explored (about 46 exploratory visits per entry), and it is excluded from the main results.

Configuration ($\gamma = 0.95$, $\alpha = 0.15$)	Δ intensity	Δ informativeness
Residual, 1.5×10^6 periods, $\beta = 4 \times 10^{-6}$	0.31 ± 0.02	0.68 ± 0.02
Residual, 6×10^6 , $\beta = 10^{-6}$, seed 0	0.88 ± 0.01	0.95 ± 0.01
Residual, 6×10^6 , $\beta = 10^{-6}$, seed 1	0.87 ± 0.01	0.93 ± 0.01
Residual, 6×10^6 , $\beta = 10^{-6}$, seed 2	0.85 ± 0.01	0.92 ± 0.01
Residual, 1.5×10^7 , $\beta = 4 \times 10^{-7}$	0.90 ± 0.01	0.97 ± 0.01
Flow, 1.5×10^6 , $\beta = 4 \times 10^{-6}$	0.66 ± 0.02	0.70 ± 0.02
Flow, 6×10^6 , $\beta = 10^{-6}$	0.67 ± 0.02	0.69 ± 0.02
No memory, 1.5×10^6 , $\beta = 4 \times 10^{-6}$	0.27 ± 0.04	0.28 ± 0.03
No memory, 6×10^6 , $\beta = 10^{-6}$ (3 seeds)	0.24 ± 0.02	0.26 ± 0.02
No memory, 1.5×10^7 , $\beta = 4 \times 10^{-7}$	0.27 ± 0.06	0.28 ± 0.05

Table 9: Robustness of the standard-Kyle outcomes. β is the exploration decay ($\varepsilon_t = e^{-\beta t}$). 100–200 markets per row; 95% CIs.

F Implementation details

Validation. The test suite (72 tests) checks that the benchmarks maximise the relevant objectives on fine grids, that the closed forms and the numerical solver agree at $\xi = 0$, and that the solver

Parameter	Value
Values v	5 (standard Kyle) or 10 (Dou replication) equiprobable normal quantiles, variance rescaled to σ_v^2
Orders	31-point grid on $\pm 1.5 \beta^N \max v $ (wide); 15 per value on the cartel-Nash bracket $\pm 10\%$
Memory bins	7 flow/residual bins on ± 2 sd; 15 price-surprise bins on ± 2.5 sd
Q-learning	$\alpha = 0.15$ (0.05 in Dou replication), $\gamma = 0.95$, $\varepsilon_t = e^{-\beta t}$
Q initialisation	discounted payoff against uniformly random rivals at λ^N
Decaying step	$\alpha_n = \alpha(1 + n/50)^{-0.7}$ per entry
Market maker	EWMA moments of (v, y) , half-life 2,000 periods, starts at Nash
Evaluation	20,000 greedy periods, learning off, market maker adapting
Deviation test	40 events per market, 50 greedy periods apart, 16-period window
Noise-shock test	shocks of 0.25 and 1 deviation units, 20 events per market
DQN	2×64 MLP, Double DQN, Huber loss, Adam 10^{-3} , buffer 5×10^4 , batch 64
PPO	2×64 actor-critic, clip 0.2, GAE $\lambda = 0.95$, 4 epochs, rollout 256, entropy annealed to 0

Table 10: Default parameters.

reproduces the Dou et al. [14] values at $\xi = 500$ ($\beta^N = 166.667$, $\beta^C = 125$, $\lambda^N = 0.002$); that simulated Nash and cartel play recover the corresponding price impact, profit and collusion index; that sessions are independent; that counterfactual profits equal realised profits for the order taken; that a resumed run reproduces an uninterrupted one exactly; and that the deviation and noise-shock tests report a reaction for scripted punishing or price-trigger traders and none for memoryless ones.